SRI (k-inflation): N01LO

The spectra are evaluated at $-k_*\eta_* = 1$, through relative amplitude order $p = 1$. The Hubble-flow parameters are denoted by ϵ_j and $M_{\rm Pl}$ is the reduced Planck mass. Coefficients are generated numerically from the normalized mode equations, using the iterative algebra. The convention for the spectral derivatives is

$$\alpha_n^A = \frac{d^{n+1}\ln P_A}{d(\ln k)^{n+1}}, \qquad \alpha_0^s = n_s - 1, \qquad \alpha_0^t = n_t, \qquad r = P_t/P_s.$$

For k-inflation, $c_t = 1$. The scalar amplitude includes transport from the acoustic reference to the common tensor reference. The sound speed is $c_{s*} > 0$, with

$$\alpha_n^{c_s} = \frac{d^{n+1}\ln c_s}{d(\ln k)^{n+1}}.$$

1 Spectra

1.1 Scalar spectrum

$$P_s(k_*) = \frac{H_*^2}{8\pi^2 M_{\rm Pl}^2 \epsilon_{1*} c_{s*}} \left[\sum_{d=0}^{1} F_{s,d} + O(\mathrm{SR}^2)\right].$$

Relative order 0.

$$F_{s,0} = 1.0$$

Relative order 1.

$$\begin{aligned} F_{s,1} = {} & 0.72963715453852\epsilon_2 - 0.54072569092296\epsilon_1 \\ & - 1.2703628454615\alpha_0^{c_s} - 1.0\alpha_0^{c_s}\log(c_s) \\ & - 1.0\epsilon_2\log(c_s) - 2.0\epsilon_1\log(c_s) \end{aligned}$$

1.2 Tensor spectrum

$$P_t(k_*) = \frac{2H_*^2}{\pi^2 M_{\rm Pl}^2} \left[\sum_{d=0}^{1} F_{t,d} + O(\mathrm{SR}^2)\right].$$

Relative order 0.

$$F_{t,0} = 1.0$$

Relative order 1.

$$F_{t,1} = -0.54072569092296\epsilon_1$$

2 Consistency relations

2.1 $n = 0$

For $p = 1$, the absolute cutoff is $W = 2$:

$$\alpha_0^t = \sum_{d=1}^{2} R_{0,d} + O(\mathrm{SR}^3).$$

Absolute order 1.

$$R_{0,1} = -\frac{0.125r}{c_s}$$

Absolute order 2.

$$R_{0,2} = \frac{0.0078125r^2}{c_s^2} + \frac{0.125\alpha_0^s r}{c_s} + \frac{0.28379535568268\alpha_0^{c_s} r}{c_s} - \frac{0.125\alpha_0^s r \log(c_s)}{c_s}$$

2.2 $n = 1$

For $p = 1$, the absolute cutoff is $W = 3$:

$$\alpha_1^t = \sum_{d=2}^{3} R_{1,d} + O(\mathrm{SR}^4).$$

Absolute order 2.

$$R_{1,2} = \frac{0.015625r^2}{c_s^2} + \frac{0.125\alpha_0^{c_s} r}{c_s} + \frac{0.125\alpha_0^s r}{c_s}$$

Absolute order 3.

$$\begin{aligned} R_{1,3} = &-\frac{0.0029296875r^3}{c_s^3} + \frac{0.125\alpha_1^s r}{c_s} \\ &+ \frac{0.28379535568268\alpha_1^{c_s} r}{c_s} - \frac{0.125\left(\alpha_0^s\right)^2 r}{c_s} \\ &- \frac{0.046875\alpha_0^s r^2}{c_s^2} - \frac{0.28379535568268\left(\alpha_0^{c_s}\right)^2 r}{c_s} \\ &- \frac{0.086573838920671\alpha_0^{c_s} r^2}{c_s^2} + \frac{0.03125\alpha_0^s r^2 \log(c_s)}{c_s^2} \\ &+ \frac{0.125\left(\alpha_0^s\right)^2 r \log(c_s)}{c_s} \\ &- \frac{0.125\alpha_1^s r \log(c_s)}{c_s} - \frac{0.53379535568268\alpha_0^{c_s}\alpha_0^s r}{c_s} \\ &+ \frac{0.125\alpha_0^{c_s}\alpha_0^s r \log(c_s)}{c_s} \end{aligned}$$

2.3 $n = 2$

For $p = 1$, the absolute cutoff is $W = 4$:

$$\alpha_2^t = \sum_{d=3}^{4} R_{2,d} + O(\mathrm{SR}^5).$$

Absolute order 3.

$$\begin{aligned} R_{2,3} = &-\frac{0.00390625r^3}{c_s^3} + \frac{0.125\alpha_1^{c_s} r}{c_s} \\ &+ \frac{0.125\alpha_1^s r}{c_s} - \frac{0.125\left(\alpha_0^{c_s}\right)^2 r}{c_s} \\ &- \frac{0.125\left(\alpha_0^s\right)^2 r}{c_s} - \frac{0.046875\alpha_0^{c_s} r^2}{c_s^2} \\ &- \frac{0.046875\alpha_0^s r^2}{c_s^2} - \frac{0.25\alpha_0^{c_s}\alpha_0^s r}{c_s} \end{aligned}$$

Absolute order 4.

$$
\begin{aligned}
R_{2,4} = & \frac{0.0013427734375 r^4}{c_s^4} + \frac{0.125 \alpha_2^s r}{c_s} \\
& + \frac{0.125 \left(\alpha_0^s\right)^3 r}{c_s} + \frac{0.125 \left(\alpha_0^s\right)^2 r^2}{c_s^2} \\
& + \frac{0.025390625 \alpha_0^s r^3}{c_s^3} + \frac{0.28379535568268 \alpha_2^{c_s} r}{c_s} \\
& + \frac{0.28379535568268 \left(\alpha_0^{c_s}\right)^3 r}{c_s} \\
& + \frac{0.24409651676201 \left(\alpha_0^{c_s}\right)^2 r^2}{c_s^2} + \frac{0.040277689595252 \alpha_0^{c_s} r^3}{c_s^3} \\
& - \frac{0.0625 \alpha_1^s r^2}{c_s^2} - \frac{0.12204825838101 \alpha_1^{c_s} r^2}{c_s^2} \\
& + \frac{0.046875 \alpha_1^s r^2 \log\left(c_s\right)}{c_s^2} \\
& + \frac{0.94259071136537 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s r}{c_s} \\
& + \frac{0.78379535568268 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 r}{c_s} \\
& + \frac{0.41597151676201 \alpha_0^{c_s} \alpha_0^s r^2}{c_s^2} - \frac{0.125 \alpha_2^s r \log\left(c_s\right)}{c_s} \\
& - \frac{0.125 \left(\alpha_0^s\right)^3 r \log\left(c_s\right)}{c_s} \\
& - \frac{0.01171875 \alpha_0^s r^3 \log\left(c_s\right)}{c_s^3} \\
& - \frac{0.09375 \left(\alpha_0^s\right)^2 r^2 \log\left(c_s\right)}{c_s^2} - \frac{0.375 \alpha_0^s \alpha_1^s r}{c_s} \\
& - \frac{0.81759071136537 \alpha_0^s \alpha_1^{c_s} r}{c_s} - \frac{0.78379535568268 \alpha_0^{c_s} \alpha_1^s r}{c_s} \\
& - \frac{0.85138606704805 \alpha_0^{c_s} \alpha_1^{c_s} r}{c_s} \\
& + \frac{0.125 \alpha_0^s \alpha_1^{c_s} r \log\left(c_s\right)}{c_s} \\
& + \frac{0.25 \alpha_0^{c_s} \alpha_1^s r \log\left(c_s\right)}{c_s} \\
& + \frac{0.375 \alpha_0^s \alpha_1^s r \log\left(c_s\right)}{c_s} \\
& - \frac{0.125 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s r \log\left(c_s\right)}{c_s} \\
& - \frac{0.25 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 r \log\left(c_s\right)}{c_s} \\
& - \frac{0.09375 \alpha_0^{c_s} \alpha_0^s r^2 \log\left(c_s\right)}{c_s^2}
\end{aligned}
$$

2.4 $n = 3$

For $p = 1$, the absolute cutoff is $W = 5$:

$$\alpha_3^t = \sum_{d=4}^{5} R_{3,d} + O(\mathrm{SR}^6).$$

Absolute order 4.

$$
\begin{aligned}
R_{3,4} = & \frac{0.00146484375 r^4}{c_s^4} + \frac{0.125 \alpha_2^{c_s} r}{c_s} \\
& + \frac{0.125 \alpha_2^s r}{c_s} + \frac{0.125 \left(\alpha_0^{c_s}\right)^3 r}{c_s} \\
& + \frac{0.125 \left(\alpha_0^s\right)^3 r}{c_s} + \frac{0.0234375 \alpha_0^{c_s} r^3}{c_s^3} \\
& + \frac{0.0234375 \alpha_0^s r^3}{c_s^3} + \frac{0.109375 \left(\alpha_0^{c_s}\right)^2 r^2}{c_s^2} \\
& + \frac{0.109375 \left(\alpha_0^s\right)^2 r^2}{c_s^2} - \frac{0.0625 \alpha_1^{c_s} r^2}{c_s^2} \\
& - \frac{0.0625 \alpha_1^s r^2}{c_s^2} + \frac{0.375 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 r}{c_s} \\
& + \frac{0.375 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s r}{c_s} + \frac{0.21875 \alpha_0^{c_s} \alpha_0^s r^2}{c_s^2} \\
& - \frac{0.375 \alpha_0^{c_s} \alpha_1^{c_s} r}{c_s} - \frac{0.375 \alpha_0^{c_s} \alpha_1^s r}{c_s} \\
& - \frac{0.375 \alpha_0^s \alpha_1^{c_s} r}{c_s} - \frac{0.375 \alpha_0^s \alpha_1^s r}{c_s}
\end{aligned}
$$

Absolute order 5.

$$
\begin{aligned}
R_{3,5} = & -\frac{0.000762939453125 r^5}{c_s^5} + \frac{0.125 \alpha_3^s r}{c_s} \\
& + \frac{0.0419921875 \alpha_1^s r^3}{c_s^3} + \frac{0.28379535568268 \alpha_3^{c_s} r}{c_s} \\
& + \frac{0.071766316690503 \alpha_1^{c_s} r^3}{c_s^3} - \frac{0.125 \left(\alpha_0^s\right)^4 r}{c_s} \\
& - \frac{0.375 \left(\alpha_1^s\right)^2 r}{c_s} - \frac{0.078125 \alpha_2^s r^2}{c_s^2} \\
& - \frac{0.28125 \left(\alpha_0^s\right)^3 r^2}{c_s^2} - \frac{0.01708984375 \alpha_0^s r^4}{c_s^4} \\
& - \frac{0.1201171875 \left(\alpha_0^s\right)^2 r^3}{c_s^3} - \frac{0.28379535568268 \left(\alpha_0^{c_s}\right)^4 r}{c_s} \\
& - \frac{0.15752267784134 \alpha_2^{c_s} r^2}{c_s^2} - \frac{0.85138606704805 \left(\alpha_1^{c_s}\right)^2 r}{c_s} \\
& - \frac{0.024533376047626 \alpha_0^{c_s} r^4}{c_s^4} - \frac{0.5591418724447 \left(\alpha_0^{c_s}\right)^3 r^2}{c_s^2} \\
& - \frac{0.20943957507151 \left(\alpha_0^{c_s}\right)^2 r^3}{c_s^3} \\
& + \frac{0.0625 \alpha_2^s r^2 \log\left(c_s\right)}{c_s^2} \\
& + \frac{0.125 \left(\alpha_0^s\right)^4 r \log\left(c_s\right)}{c_s} \\
& + \frac{0.005859375 \alpha_0^s r^4 \log\left(c_s\right)}{c_s^4} \\
& + \frac{0.375 \left(\alpha_1^s\right)^2 r \log\left(c_s\right)}{c_s} \\
& + \frac{0.75 \left(\alpha_0^s\right)^2 \alpha_1^s r}{c_s}
\end{aligned}
$$

$$
\begin{aligned}
&+ \frac{0.21875\left(\alpha_0^s\right)^3 r^2 \log\left(c_s\right)}{c_s^2} \\
&+ \frac{0.4375\alpha_0^s\alpha_1^s r^2}{c_s^2} \\
&+ \frac{0.0703125\left(\alpha_0^s\right)^2 r^3 \log\left(c_s\right)}{c_s^3} \\
&+ \frac{0.72129535568268\alpha_0^{c_s}\alpha_1^s r^2}{c_s^2} \\
&+ \frac{1.7027721340961\left(\alpha_0^{c_s}\right)^2 \alpha_1^{c_s} r}{c_s} \\
&+ \frac{1.6013860670481\left(\alpha_0^s\right)^2 \alpha_1^{c_s} r}{c_s} \\
&+ \frac{1.9763860670481\left(\alpha_0^{c_s}\right)^2 \alpha_1^s r}{c_s} \\
&+ \frac{0.7778918724447\alpha_0^s\alpha_1^{c_s} r^2}{c_s^2} \\
&+ \frac{0.87418722812738\alpha_0^{c_s}\alpha_1^{c_s} r^2}{c_s^2} - \frac{0.125\alpha_3^s r \log\left(c_s\right)}{c_s} \\
&- \frac{0.5\alpha_0^s\alpha_2^s r}{c_s} - \frac{0.0234375\alpha_1^s r^3 \log\left(c_s\right)}{c_s^3} \\
&- \frac{1.1351814227307\alpha_0^{c_s}\alpha_2^{c_s} r}{c_s} - \frac{1.0337953556827\alpha_0^{c_s}\alpha_2^s r}{c_s} \\
&- \frac{1.0337953556827\alpha_0^{c_s}\left(\alpha_0^s\right)^3 r}{c_s} \\
&- \frac{1.1013860670481\alpha_0^s\alpha_2^{c_s} r}{c_s} \\
&- \frac{1.3513860670481\left(\alpha_0^{c_s}\right)^3 \alpha_0^s r}{c_s} \\
&- \frac{1.6013860670481\alpha_1^{c_s}\alpha_1^s r}{c_s} \\
&- \frac{1.9763860670481\left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^2 r}{c_s} \\
&- \frac{1.6182837448894\left(\alpha_0^{c_s}\right)^2 \alpha_0^s r^2}{c_s^2} \\
&- \frac{1.3403918724447\alpha_0^{c_s}\left(\alpha_0^s\right)^2 r^2}{c_s^2} \\
&- \frac{0.35299426257151\alpha_0^{c_s}\alpha_0^s r^3}{c_s^3} \\
&+ \frac{0.125\alpha_0^s\alpha_2^{c_s} r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.125\left(\alpha_0^{c_s}\right)^3 \alpha_0^s r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.5\alpha_0^s\alpha_2^s r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.375\alpha_0^{c_s}\alpha_2^s r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.375\alpha_0^{c_s}\left(\alpha_0^s\right)^3 r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.375\alpha_1^{c_s}\alpha_1^s r \log\left(c_s\right)}{c_s}
\end{aligned}
$$

$$
\begin{aligned}
&+ \frac{0.375 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^2 r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.21875 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s r^2 \log\left(c_s\right)}{c_s^2} \\
&+ \frac{0.4375 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 r^2 \log\left(c_s\right)}{c_s^2} \\
&+ \frac{0.0703125 \alpha_0^{c_s} \alpha_0^s r^3 \log\left(c_s\right)}{c_s^3} \\
&+ \frac{3.1013860670481 \alpha_0^{c_s} \alpha_0^s \alpha_1^s r}{c_s} \\
&+ \frac{3.6791582011442 \alpha_0^{c_s} \alpha_0^s \alpha_1^{c_s} r}{c_s} \\
&- \frac{0.125 \alpha_0^s \alpha_1^{c_s} r^2 \log\left(c_s\right)}{c_s^2} \\
&- \frac{0.375 \left(\alpha_0^s\right)^2 \alpha_1^{c_s} r \log\left(c_s\right)}{c_s} \\
&- \frac{0.375 \left(\alpha_0^{c_s}\right)^2 \alpha_1^s r \log\left(c_s\right)}{c_s} \\
&- \frac{0.75 \left(\alpha_0^s\right)^2 \alpha_1^s r \log\left(c_s\right)}{c_s} \\
&- \frac{0.21875 \alpha_0^{c_s} \alpha_1^s r^2 \log\left(c_s\right)}{c_s^2} \\
&- \frac{0.34375 \alpha_0^s \alpha_1^s r^2 \log\left(c_s\right)}{c_s^2} \\
&- \frac{0.375 \alpha_0^{c_s} \alpha_0^s \alpha_1^{c_s} r \log\left(c_s\right)}{c_s} \\
&- \frac{1.125 \alpha_0^{c_s} \alpha_0^s \alpha_1^s r \log\left(c_s\right)}{c_s}
\end{aligned}
$$